

EDUCATION

Carnegie Mellon University
Bachelor of Science in Computer Science, Minor in Machine Learning
Coursework: ML (PhD), Deep Learning Systems (MS), Distributed Systems, Computer Systems, Parallel DSA
Honors/Activities: SCS Dean’s List, CMU AI Safety Initiative, Scotty Labs

Pittsburgh, PA
Dec 2026

WORK EXPERIENCE

Notion | Software Engineer Intern
May 2026 – Present

- Developing a multi-agent system to autonomously scan and patch Notion infrastructure, fixing 57 vulnerabilities to date
- Created agent-driven tooling to track, rotate, and deploy 1300+ secrets across ECS, K8s, Lambda, and CI/CD targets
- Built proxy infrastructure for sandboxed AI agents, resolving placeholders into sensitive runtime data and sanitizing I/O

Tesla (Autopilot) | Software Engineer Intern
Jan 2026 - May 2026

- Built isolated, performance-optimized Autopilot infrastructure for production model builds, resolving a P0 security directive
- Improved model encryption memory footprint by 98%, and reduced throttling during Autopilot simulation runs by 47%
- Accelerated ML compilation and testing pipeline with weights deduplication (-25% build volume), SDK caching (-20% time)
- Migrated 34 services such as orchestrators for inference clusters, simulation, replay across K8s, VM, baremetal to new DCs

Amazon Web Services | Software Engineer Intern, AI/ML
May 2025 – August 2025

- Built service within AWS Agentic AI to translate Bedrock Agents into equivalent agents in other frameworks (eg. Langchain)
- Recreated 20+ Bedrock primitives including guardrails, Lambda tools, memory, agent collaboration, and code sandboxing
- Achieved 95% agent similarity with continuous eval pipeline on feature-conditional prompt suite and 14k randomized agents
- Featured on AWS Console , launched with SDK integration , and validated with clients including Autodesk and Netflix

Hadrian | Software Engineer Intern
May 2024 – Aug 2024

- Developed Python/Go backend and Kafka pipeline (<50ms latency) to enable robotic tool-loading and sync with factory state
- Trained and deployed on-device ML model achieving 98.5% accuracy in fully automated validation of tool position/type
- Cut tool-loading times by 75%, saving 11k technician-hours yearly in building parts for clients such as Anduril and Raytheon

Compos Mentis | Founding Engineer
Aug 2023 – May 2024

- Developed platform using LLMs to accelerate case qualification and legal review, generating \$90k revenue from 23 law firms
- Fine-tuned LLMs via LoRA and built extraction and RAG pipeline to surface case insights, speeding up analysis by 8x

Machina Labs | Software Development Intern
Jul 2022 – Sep 2022

- Created ML pipeline to improve part alignment using laser scans and Bayesian optimization; deployed with Azure Kubernetes
- Achieved 40% faster iteration and 25% reduced margin of error on parts for clients such as NASA, SpaceX, and the USAF

PROJECTS

Quantum NNs CUDA ops, QNN modules for autograd-based DL framework , beating classical NNs by up to 42%

Train of Thought Multi-agent LLM system capable of chain-of-thought reasoning, backtracking with Google Gemini

TexPilot Extension for context-aware LaTeX code completions and explanations with LLaMA 2

File-Caching Proxy Low-latency distributed system for remote file operations using a server and caching proxy

PDD Model ResNet-9 model to identify 38 plant diseases with 93% accuracy, deployed with AWS Sagemaker

ACHIEVEMENTS

VEX Robotics World Championship (2x) | Achieved top ~30 out of 11k teams as Team Captain (2019-23)

AWS Certified Solutions Architect | Lockheed Martin Scholar (\$40k) | Edison International STEM Scholar (\$50k)

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, Java, Go, SQL, Bash, HTML/CSS

Services: AWS, Azure, GCP, Kubernetes, Airflow, Spark, Elasticsearch, Kafka, Redis, MySQL, PostgreSQL

Tools/Frameworks: Git, Docker, Node.js, TensorFlow, PyTorch, CUDA, NumPy, React.js, Express, Flask, gRPC